

Lynbrook Math Summer Camp 2026

A2 Number Sense

Lynbrook Math Club

August 2026

A2.1 Lecture

What is Number Sense?

Computations appear throughout mathematics, from basic arithmetic to complex algebraic manipulations. Although their roles vary across different branches of mathematics, one principle remains consistent: recognizing patterns and making insightful observations often leads to faster and more elegant solutions. Number sense is a collection of these intuitive skills. Rather than relying on brute force, strong number sense allows us to manipulate numbers efficiently, estimate quickly, and make connections. It is one of the most important skills to master for success in mathematics competitions.

For example, consider the product

$$53 \cdot 47.$$

Instead of calculating directly, we might notice that the two numbers are equally spaced around 50, a number that is familiar and easy to work with. This motivates us to use the difference of squares:

$$\begin{aligned} 53 \cdot 47 &= (50 + 3)(50 - 3) \\ &= 50^2 - 3^2 \\ &= 2491. \end{aligned}$$

By doing this, we have transformed what initially seems like a tedious calculation into a much simpler one. The goal of this lecture is to develop this intuition and strengthen our ability to recognize useful patterns.

Throughout this chapter, we will introduce techniques for simplifying arithmetic and evaluating expressions efficiently. The aim is not to teach specialized “tricks”, but rather to foster an understanding of why these methods work and how they can be applied to unfamiliar problems.

General Multiplications

Multiplication is an important part of number sense, so it is natural to seek a method to multiply two numbers quickly and accurately. Multiplication of two one-digit numbers is straightforward, but products involving larger numbers can seem more intimidating. To address this, we introduce a structured multiplication method based on expansion.

Consider two two-digit numbers $\overline{ab}$ and $\overline{cd}$. Rather than treating them as whole numbers, we can write them in expanded form:

$$\begin{aligned}\overline{ab} \cdot \overline{cd} &= (10a + b)(10c + d) \\ &= 100ac + 10(ad + bc) + bd.\end{aligned}$$

The expansion reveals that two two-digit numbers can be multiplied by combining several smaller products involving their individual digits. In particular, we can track the units, tens, and hundreds place contributions separately.

As an example, suppose we wish to compute $47 \cdot 93$. The calculation can be visualized as follows:

$$\begin{array}{ccc} 47 & 47 & 47 \\ \updownarrow & \boxtimes & \updownarrow \\ 93 & 93 & 93 \end{array}$$

We calculate each place value contribution independently:

$$\begin{aligned}\text{Units: } 3 \cdot 7 &= \mathbf{21} \\ \text{Tens: } 4 \cdot 3 + 7 \cdot 9 + \mathbf{2} &= \mathbf{77} \\ \text{Hundreds: } 4 \cdot 9 + 7 &= \mathbf{43}.\end{aligned}$$

Combining these results yields $47 \cdot 93 = 4371$.

The same idea extends naturally to three-digit numbers and beyond. Though the number of intermediate products increases, the principle behind the method remains consistent.

Powers of Ten in Multiplication

In addition to general multiplication techniques, it is helpful to recognize certain numbers with especially useful properties.

When performing computations, one should identify numbers that divide powers of 10. The motivation is that it is considerably more straightforward to multiply by powers of 10 than by most other integers. For example, since

$$25 = \frac{100}{4},$$

we can compute

$$\begin{aligned}41 \cdot 25 &= 41 \cdot \frac{100}{4} \\ &= 10.25 \cdot 100 \\ &= 1025.\end{aligned}$$

More generally, when a product involves a divisor d of 10^k , it is often advantageous to express d as

$$\frac{10^k}{d}$$

so that the multiplication can be performed by first dividing by $\frac{10^k}{d}$ and then multiplying by 10^k .

The usefulness of powers of 10 extends beyond exact computation. They can also be used to construct convenient approximations. For instance,

$$226 \approx \frac{9}{4} \cdot 100.$$

This substitution is likely to make subsequent calculations simpler while introducing only a negligible error.

More Multiplication Techniques

This section contains a collection of identities that may be useful for multiplication problems.

The first technique is essentially a way of multiplying by 1 in disguise: we double one number and halve the other (note that this preserves the product).

$$\begin{aligned} 45 \cdot 68 &= (45 \cdot 2) \cdot \frac{68}{2} \\ &= 90 \cdot 34 \\ &= 3060. \end{aligned}$$

Another example:

$$\begin{aligned} 18 \cdot 37 &= 9 \cdot 74 \\ &= 666. \end{aligned}$$

This next technique is motivated by the desire to take advantage of convenient numbers. In particular, one can write numbers in terms of an adjacent multiple of 10. Here is an example:

$$\begin{aligned} 46 \cdot 53 &= (50 - 4)(50 + 3) \\ &= 50^2 + 3 \cdot 50 - 4 \cdot 50 - 4 \cdot 3 \\ &= 2438. \end{aligned}$$

Another example:

$$\begin{aligned} 87 \cdot 104 &= (90 - 3)(100 + 4) \\ &= 9000 + 360 - 300 - 12 \\ &= 9048. \end{aligned}$$

This final shortcut can be used when two numbers are equidistant from a convenient number. This is an application of the difference of squares.

$$\begin{aligned} 46 \cdot 34 &= (40 + 6)(40 - 6) \\ &= 40^2 - 6^2 \\ &= 1564. \end{aligned}$$

Note that this is a special case of the previous idea.

Perfect Squares

It is always useful to be able to compute perfect squares quickly.

To square a number with units digit 0, note that we can factor out powers of 10:

$$\begin{aligned} 90^2 &= 9^2 \cdot 10^2 \\ &= 8100. \end{aligned}$$

For integers ending in 5,

$$\begin{aligned} (10a + 5)^2 &= 100a^2 + 100a + 25 \\ &= 100a(a + 1) + 25. \end{aligned}$$

Informally, we multiply a by $a + 1$ and append 25. For example,

$$\begin{aligned} 135^2 &= 100(13 \cdot 14) + 25 \\ &= 18225. \end{aligned}$$

We can compute squares of other integers by using a difference of squares. Consider the following example:

$$\begin{aligned} 56^2 - 4^2 &= (56 - 4)(56 + 4) \implies 56^2 = 52 \cdot 60 + 4^2 \\ &\implies 56^2 = 3136. \end{aligned}$$

The advantage of this method is that one only needs to memorize the squares of the one-digit numbers. Note that one should choose the square to subtract such that the resulting product is convenient to calculate. Suppose that we subtract 5^2 instead of 4^2 :

$$\begin{aligned} 56^2 - 5^2 &= (56 - 5)(56 + 5) \implies 56^2 = 51 \cdot 61 + 5^2 \\ &\implies 56^2 = 3136. \end{aligned}$$

Although the result is the same (as it should be), the computation in the second example takes a fair bit longer.

Conclusion

The techniques presented in this chapter can greatly simplify mental calculations, but they are most effective when they become second nature through consistent practice. As stated previously, the ultimate goal is to develop an intuition for numbers and the relationships between them. With practice and experience, these techniques become increasingly natural and can often be applied instinctively.